First Project with Open-RMF

Creating ROS packages

Create a ROS workspace

```
mkdir -p /home/usuario/rmf_ws/src
```

Launch terminal in Open-RMF container

```
cd /home/usuario

rocker --nvidia --x11 --name rmf_demos \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  --network host --user \
  --volume `pwd`/rmf_ws:/home/usuario/rmf_ws --\
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    bash
```

Check the office simulation

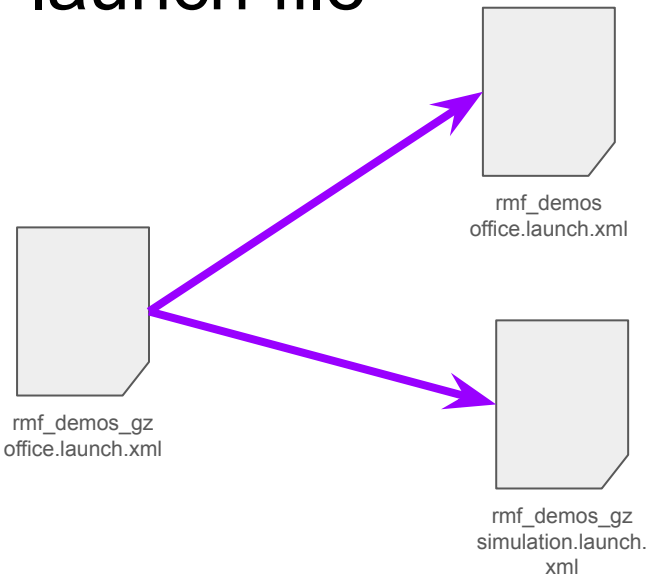
```
sudo cp -R /root/.gazebo .
ros2 launch rmf_demos_gz \
  office.launch.xml
```

Task: create a ROS package for the first project

Create a ROS package for the first project with all the necessary files for building the world and launching the simulation with a fleet of Tinyrobots.

To do so, let’s examine the files used in the simulation of the office world of the `rmf_demos` package.

Structure of the RMF office launch file




```
<?xml version='1.0' ?>

<launch>
  <arg name="use_sim_time" default="true"/>
  <arg name="failover_mode" default="false"/>
  <arg name="gazebo_version" default='8'/>
  <arg name="sim_update_rate" default='100'/>

  <!-- Common launch -->
  <include file="$(find-pkg-share rmf_demos)/office.launch.xml">
    <arg name="use_sim_time" value="$(var use_sim_time)"/>
    <arg name="failover_mode" value="$(var failover_mode)"/>
  </include>

  <!-- Simulation launch -->
  <include file="$(find-pkg-share rmf_demos_gz)/simulation.launch.xml">
    <arg name="map_name" value="office" />
    <arg name="gazebo_version" value="$(var gazebo_version)" />
    <arg name="sim_update_rate" value="$(var sim_update_rate)"/>
  </include>

</launch>
```

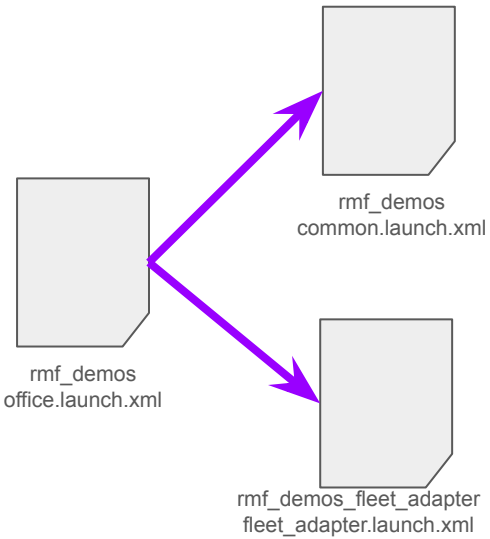
https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos_gz**/launch/**office.launch.xml**

```xml
<?xml version='1.0' ?>

<launch>
  <arg name="map_package" default="rmf_demos_maps" description="Name of the map package" />
  <arg name="map_name" description="Name of the rmf_demos map to simulate" />
  <arg name="use_crowdsim" default='0'/>
  <arg name="gazebo_version" default='8'/>
  <arg name="sim_update_rate" default='100'/>

  <let name="world_path" value="$(find-pkg-share $(var map_package))/maps/$(var map_name)/$(var map_name).world" />
  <let name="model_path" value="$(find-pkg-share $(var map_package))/maps/$(var map_name)/models:$(env HOME)/.gazebo/models" />

  <!-- Use crowd sim if `use_crowdsim` is true-->
  <let name="menge_resource_path" if="$(var use_crowdsim)" value="$(find-pkg-share $(var map_package))/maps/$(var map_name)/config_resource"/>
  <let name="menge_resource_path" unless="$(var use_crowdsim)" value="" />

  <let name="gz_headless" if="$(var headless)" value="-s"/>
  <let name="gz_headless" unless="$(var headless)" value="" />

  <!-- TODO(luca) Remove the manual concatenation of GZ_SIM_RESOURCE_PATH and just use environment hooks -->
  <executable cmd="gz sim --force-version $(var gazebo_version) $(var gz_headless) -r -v 3 $(var world_path) -z $(var sim_update_rate)" output="both">
    <env name="GZ_SIM_RESOURCE_PATH" value="$(env GZ_SIM_RESOURCE_PATH):$(var model_path)" />
    <env name="MENGE_RESOURCE_PATH" value="$(var menge_resource_path)"/>
  </executable>

  <!-- ros_gz bridge for simulation clock -->
  <node pkg="ros_gz_bridge" exec="parameter_bridge"
    args="/clock@rosgraph_msgs/msg/Clock[gz.msgs.Clock"
    />

</launch>
```

Simulation launch

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos_gz**/launch/**simulation.launch.xml**

Common launch (i)




```xml
<?xml version='1.0' ?>

<launch>
  <arg name="use_sim_time" default="false"/>

  <!-- Common launch -->
  <include file="$(find-pkg-share rmf_demos)/common.launch.xml">
    <arg name="use_sim_time" value="$(var use_sim_time)"/>
    <arg name="viz_config_file" value ="$(find-pkg-share rmf_demos)/include/office/office.rviz"/>
    <arg name="config_file" value="$(find-pkg-share rmf_demos_maps)/office/office.building.yaml"/>
    <arg name="use_reservation_node" value="true"/>
  </include>

  <!-- TinyRobot fleet adapter -->
  <group>
    <include file="$(find-pkg-share rmf_demos_fleet_adapter)/launch/fleet_adapter.launch.xml">
      <arg name="use_sim_time" value="$(var use_sim_time)"/>
      <arg name="nav_graph_file" value="$(find-pkg-share rmf_demos_maps)/maps/office/nav_graphs/0.yaml" />
      <arg name="config_file" value="$(find-pkg-share rmf_demos)/config/office/tinyRobot_config.yaml"/>
    </include>
  </group>

</launch>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos**/launch/**office.launch.xml**

Common launch (ii)

```xml
<?xml version='1.0' ?>

<launch>

  <arg name="use_sim_time" default="false" description="Use the /clock topic for time to sync with simulation"/>
  <arg name="viz_config_file" default="$(find-pkg-share rmf_visualization_schedule)/config/rmf.rviz"/>
  <arg name="config_file" description="Building description file required by rmf_building_map_tools"/>
  <arg name="initial_map" default="L1" description="Initial map name for the visualizer"/>
  <arg name="headless" default="false" description="do not launch rviz; launch gazebo in headless mode"/>
  <arg name="bidding_time_window" description="Time window in seconds for task bidding process" default="2.0"/>
  <arg name="use_unique_hex_string_with_task_id" default="true" description="Appends a unique hex string to the task ID"/>
  <arg name="server_uri" default="" description="The URI of the api server to transmit state and task information."/>
  <arg name="use_reservation_node" default="false" description="Enable the reservation node for parking spot de-confliction"/>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos**/launch/**common.launch.xml**

Common launch (iii)

```
<!-- Traffic Schedule  -->
<node pkg="rmf_traffic_ros2" exec="rmf_traffic_schedule" output="both" name="rmf_traffic_schedule_primary">
  <param name="use_sim_time" value="$(var use_sim_time)"/>
</node>

<!-- Blockade Moderator -->
<node pkg="rmf_traffic_ros2" exec="rmf_traffic_blockade" output="both">
  <param name="use_sim_time" value="$(var use_sim_time)"/>
</node>

<!-- Building Map  -->
<group>
  <node pkg="rmf_building_map_tools" exec="building_map_server" args="$(var config_file)">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
  </node>
</group>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos**/launch/**common.launch.xml**

Common launch (iv)

```
<!-- Visualizer -->
<group>
  <include file="$(find-pkg-share rmf_visualization)/visualization.launch.xml">
    <arg name="use_sim_time" value="$(var use_sim_time)"/>
    <arg name="map_name" value="$(var initial_map)"/>
    <arg name="viz_config_file" value ="$(var viz_config_file)"/>
    <arg name="headless" value="$(var headless)"/>
  </include>
</group>

<!-- Door Supervisor -->
<group>
  <node pkg="rmf_fleet_adapter" exec="door_supervisor">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
  </node>
</group>

<!-- Lift Supervisor -->
<group>
  <node pkg="rmf_fleet_adapter" exec="lift_supervisor">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
  </node>
</group>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos**/launch/**common.launch.xml**

Common launch (v)

```
  <!-- Dispatcher Node -->
  <group>
    <node pkg="rmf_task_ros2" exec="rmf_task_dispatcher"  output="screen">
      <param name="use_sim_time" value="$(var use_sim_time)"/>
      <param name="bidding_time_window" value="$(var bidding_time_window)"/>
      <param name="use_unique_hex_string_with_task_id" value="$(var use_unique_hex_string_with_task_id)"/>
      <param name="server_uri" value="$(var server_uri)"/>
    </node>
  </group>

  <!-- Proto-Reservation system-->
  <group  if="$(var use_reservation_node)">
    <node pkg="rmf_reservation_node" exec="queue_manager"></node>
  </group>

</launch>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos**/launch/**common.launch.xml**

Visualization (i)

```
<?xml version = '1.0' ?>

<launch>

  <arg name="use_sim_time" default="false"/>
  <arg name="rate" default="10.0" description= "The rate in hz at which the scheduler markers should publish"/>
  <arg name="map_name" default="B1" description="The name of the map to initialize the visualizer"/>
  <arg name="viz_config_file" default="$(find-pkg-share rmf_visualization_schedule)/config/rmf.rviz"/>
  <arg name="display_names" default="true"/>
  <arg name="websocket_port" default="8006"/>
  <arg name="headless" default="false" description="Do not launch rviz2"/>
  <arg name="path_width" default="0.2" description="Width of schedule path(m) to be rendered"/>
  <arg name="lane_width" default="0.5" description="Width of lanes(m) to be rendered"/>
  <arg name="waypoint_scale" default="1.3" description="The relative size of a waypoint wrt the lane width"/>
  <arg name="text_scale" default="0.7" description="The relative size of a text label wrt the lane width"/>
  <arg name="lane_transparency" default="0.6" description="The transparency of a rendered lane in default state"/>
  <arg name="wait_secs" default="10" description="The nummber of seconds the schedule visualizer should keep trying to connect to the rmf_schedule_node"/>
  <arg name="retained_history_count" default="50" description="The retained history count for negotiations"/>
  <arg name="fleet_state_nose_scale" default="0.5" description="The radius of the nose marker as a fraction of the fleet's footprint radius"/>
  <arg name="rmf_frame_id" default="map" description="The RMF global frame id"/>
```

https://github.com/open-rmf/rmf_visualization/blob/main/**rmf_visualization**/launch/**visualization.launch.xml**

Visualization (ii)

```
<group>
  <node pkg="rmf_visualization_schedule" exec="schedule_visualizer_node">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
    <param name="rate" value="$(var rate)"/>
    <param name="path_width" value="$(var path_width)"/>
    <param name="initial_map_name" value="$(var map_name)"/>
    <param name="wait_secs" value="$(var wait_secs)"/>
    <param name="port" value="$(var websocket_port)"/>
    <param name="retained_history_count" value="$(var retained_history_count)"/>
  </node>

  <node pkg="rmf_visualization_fleet_states" exec="fleetstates_visualizer_node">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
    <param name="fleet_state_nose_scale" value="$(var fleet_state_nose_scale)"/>
    <!-- Radius parameters for fleets available in rmf_demos -->
    <param name="tinyRobot_radius" value="0.3"/>
    <param name="deliveryRobot_radius" value="0.6"/>
    <param name="cleanerBotA_radius" value="1.0"/>
    <param name="cleanerBotE_radius" value="1.0"/>
    <param name="caddy_radius" value="1.5"/>
  </node>
```

https://github.com/open-rmf/rmf_visualization/blob/main/**rmf_visualization**/launch/**visualization.launch.xml**

Visualization (iii)

```
<node pkg="rmf_visualization_building_systems" exec="rmf_visualization_building_systems" args="-m $(var map_name)">
  <param name="use_sim_time" value="$(var use_sim_time)"/>
</node>

<node pkg="rmf_visualization_navgraphs" exec="navgraph_visualizer_node">
  <param name="use_sim_time" value="$(var use_sim_time)"/>
  <param name="initial_map_name" value="$(var map_name)"/>
  <param name="lane_width" value="$(var lane_width)"/>
  <param name="waypoint_scale" value="$(var waypoint_scale)"/>
  <param name="text_scale" value="$(var text_scale)"/>
  <param name="lane_transparency" value="$(var lane_transparency)"/>
</node>
```

https://github.com/open-rmf/rmf_visualization/blob/main/**rmf_visualization**/launch/**visualization.launch.xml**

Visualization (iv)

```
  <node pkg="rmf_visualization_obstacles" exec="obstacle_visualizer_node">
    <param name="use_sim_time" value="$(var use_sim_time)"/>
    <param name="initial_map_name" value="$(var map_name)"/>
    <param name="global_fixed_frame" value="$(var rmf_frame_id)"/>
  </node>
</group>

<group unless="$(var headless)">
  <executable cmd="rviz2 -d $(var viz_config_file)" output="both"/>
</group>
```

https://github.com/open-rmf/rmf_visualization/blob/main/**rmf_visualization**/launch/**visualization.launch.xml**

Fleet adapter launch (i)

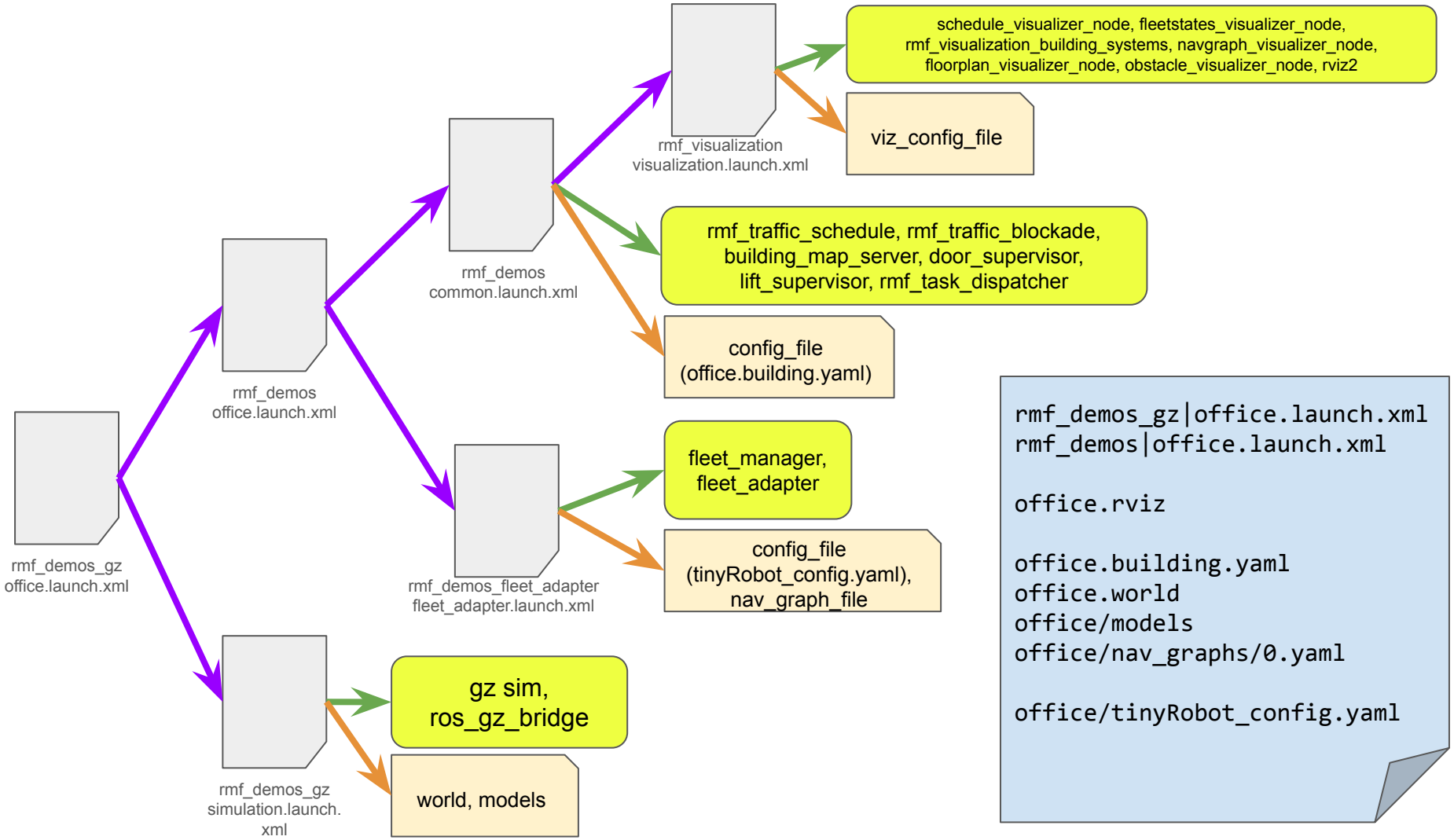
```
<?xml version='1.0' ?>

<launch>

  <arg name="use_sim_time" default="true" description="Use the /clock topic for time to sync with simulation"/>
  <arg name="config_file" description="The config file that provides important parameters for setting up the adapter"/>
  <arg name="nav_graph_file" description="The graph that this fleet should use for navigation"/>
  <arg name="server_uri" default="" description="The URI of the api server to transmit state and task information."/>
  <arg name="output" default="screen"/>

  <!-- Fleet manager -->
  <node pkg="rmf_demos_fleet_adapter"
        exec="fleet_manager"
        args="--config_file $(var config_file)"
        output="both">

    <param name="use_sim_time" value="$(var use_sim_time)"/>
  </node>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos_fleet_adapter**/launch/**fleet_adapter.launch.xml**

Fleet adapter launch (ii)

```
<!-- Fleet adapter -->
<group>
  <group if="$(var use_sim_time)">
    <node pkg="rmf_demos_fleet_adapter"
          exec="fleet_adapter"
          args="-c $(var config_file) -n $(var nav_graph_file) -sim"
          output="both">

      <param name="use_sim_time" value="$(var use_sim_time)"/>
      <param name="server_uri" value="$(var server_uri)"/>
    </node>
  </group>

  <group unless="$(var use_sim_time)">
    <node pkg="rmf_demos_fleet_adapter"
          exec="fleet_adapter"
          args="-c $(var config_file) -n $(var nav_graph_file)"
          output="both">

      <param name="use_sim_time" value="$(var use_sim_time)"/>
      <param name="server_uri" value="$(var server_uri)"/>
    </node>
  </group>
</group>

</launch>
```

https://github.com/open-rmf/rmf_demos/blob/main/**rmf_demos_fleet_adapter**/launch/**fleet_adapter.launch.xml**


rmf_demos_gz
office.launch.xml
rmf_demos
office.launch.xml
rmf_demos
common.launch.xml
rmf_visualization
visualization.launch.xml
schedule_visualizer_node, fleetstates_visualizer_node,
rmf_visualization_building_systems, navgraph_visualizer_node,
floorplan_visualizer_node, obstacle_visualizer_node, rviz2
viz_config_file
rmf_traffic_schedule, rmf_traffic_blockade,
building_map_server, door_supervisor,
lift_supervisor, rmf_task_dispatcher
config_file
(office.building.yaml)
rmf_demos_fleet_adapter
fleet_adapter.launch.xml
fleet_manager,
fleet_adapter
config_file
(tinyRobot_config.yaml),
nav_graph_file
rmf_demos_gz
simulation.launch.
xml
gz sim,
ros_gz_bridge
world, models
rmf_demos_gz|office.launch.xml
rmf_demos|office.launch.xml
office.rviz
office.building.yaml
office.world
office/models
office/nav_graphs/0.yaml
office/tinyRobot_config.yaml

Packages and files

```
rmf_demos_gz        launch/office.launch.xml
rmf_demos           launch/office.launch.xml
rmf_demos           launch/include/office/office.rviz
rmf_demos_maps      maps/office/office.building.yaml
                    share/rmf_demos_maps/maps/office/office.world
          install   share/rmf_demos_maps/maps/office/models
                    share/rmf_demos_maps/maps/office/nav_graphs/0.yaml
rmf_demos           config/office/tinyRobot_config.yaml
```

Exercise

Create a ROS package with a new Open-RMF world, including a robot.

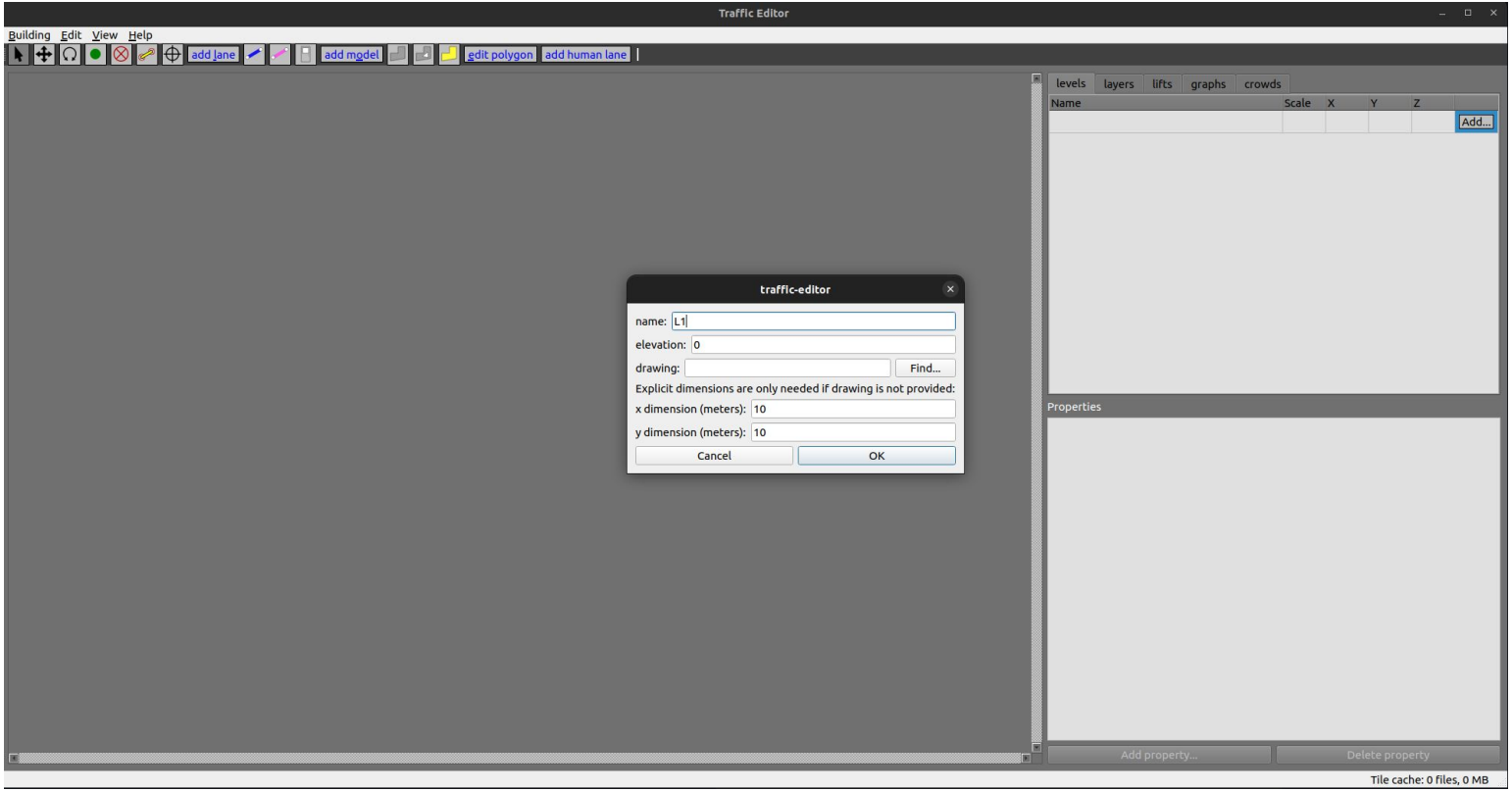
First create a simple world according to the following slides.

Then create a second world with the ICC Kyoto model including robots.

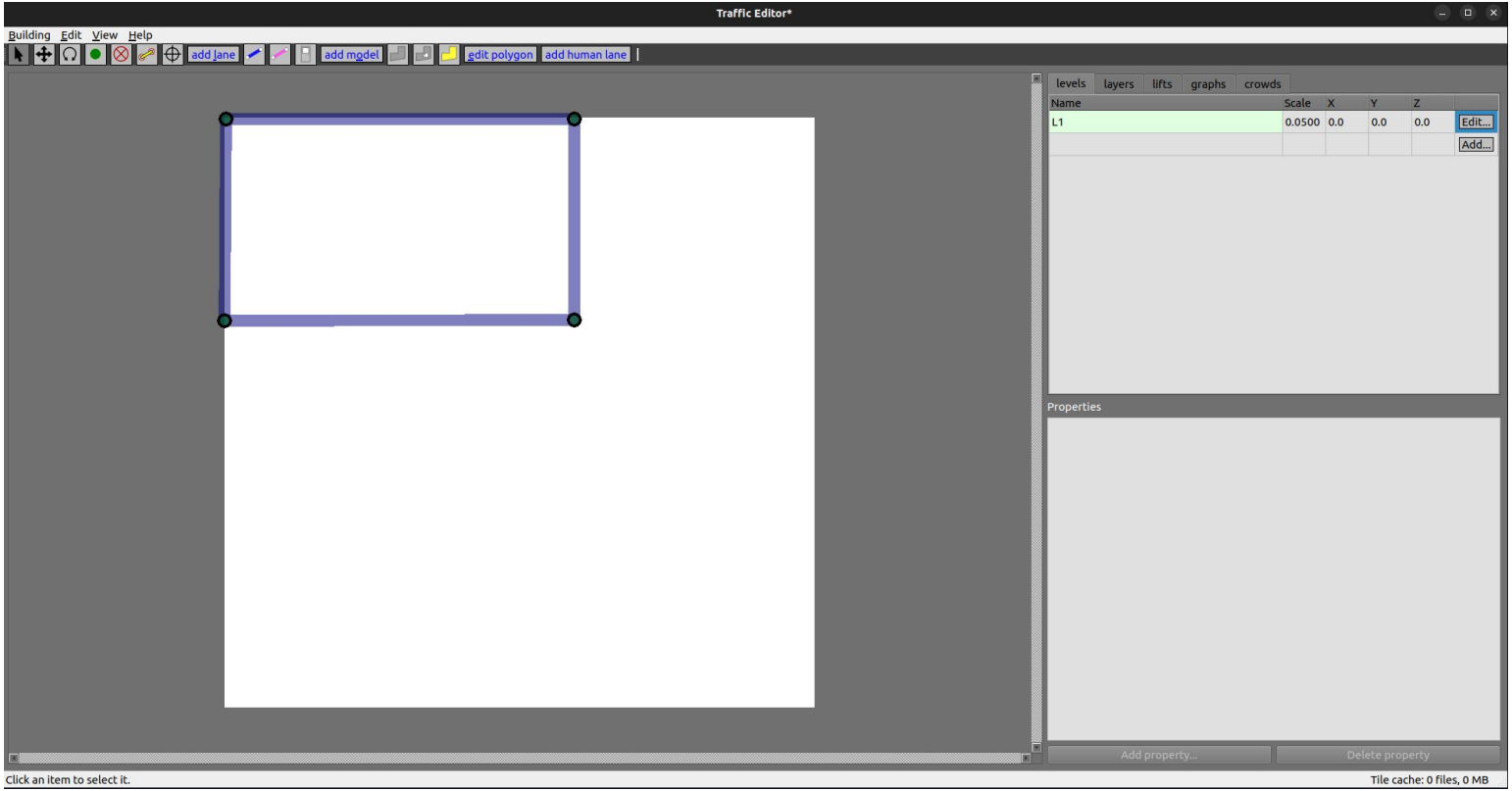
Refer to the RMF_demos packages for building the models, and creating all the necessary launch files.

RUA: Practical guide to implement OpenRMF in multi-robot environments

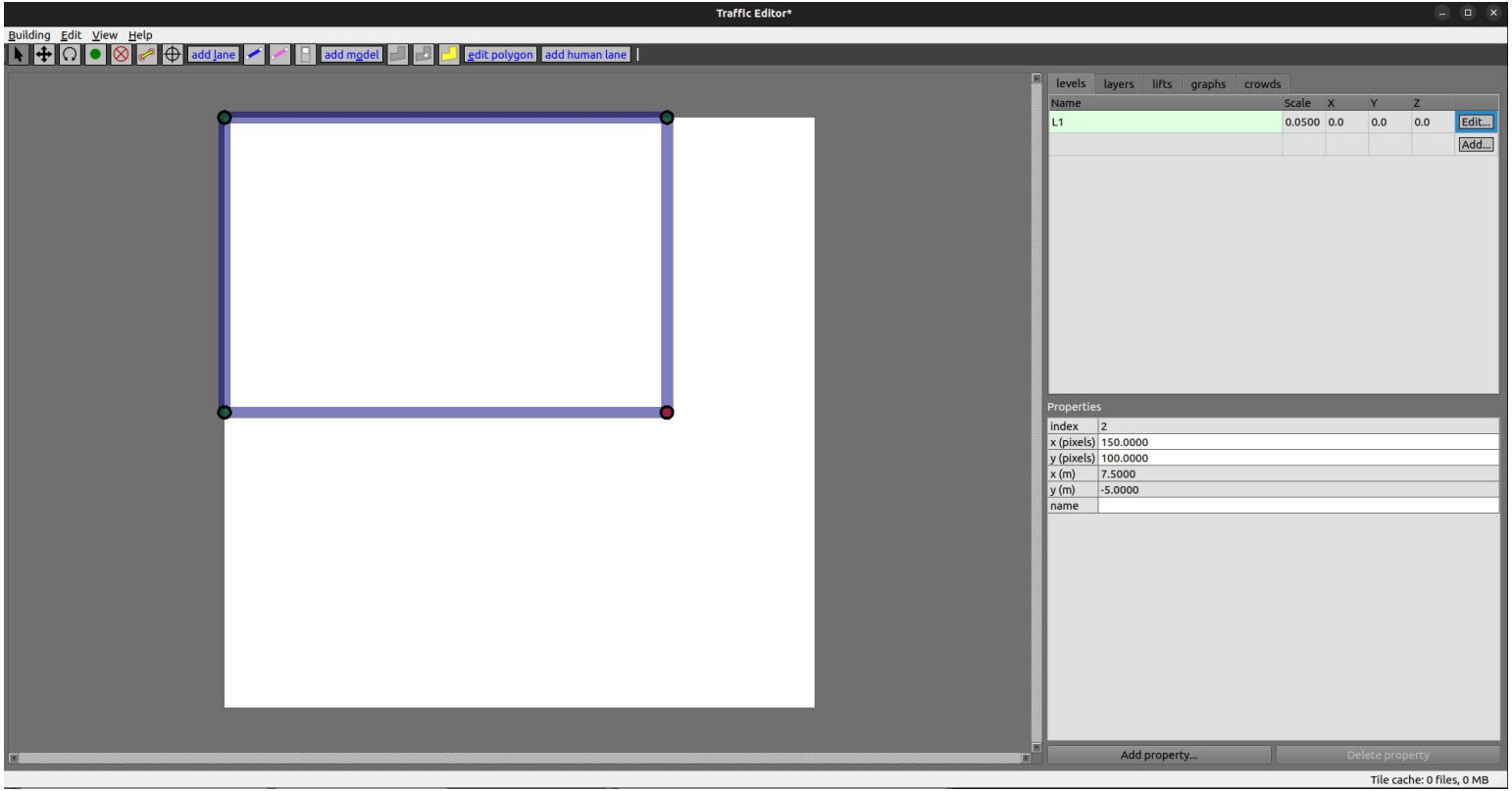
Create a map


Traffic Editor
Building Edit View Help
add lane
add model
edit polygon
add human lane
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
Add...
Properties
Add property...
Delete property
Tile cache: 0 files, 0 MB
traffic-editor
name: L1
elevation: 0
drawing:
Find...
Explicit dimensions are only needed if drawing is not provided:
x dimension (meters): 10
y dimension (meters): 10
Cancel
OK

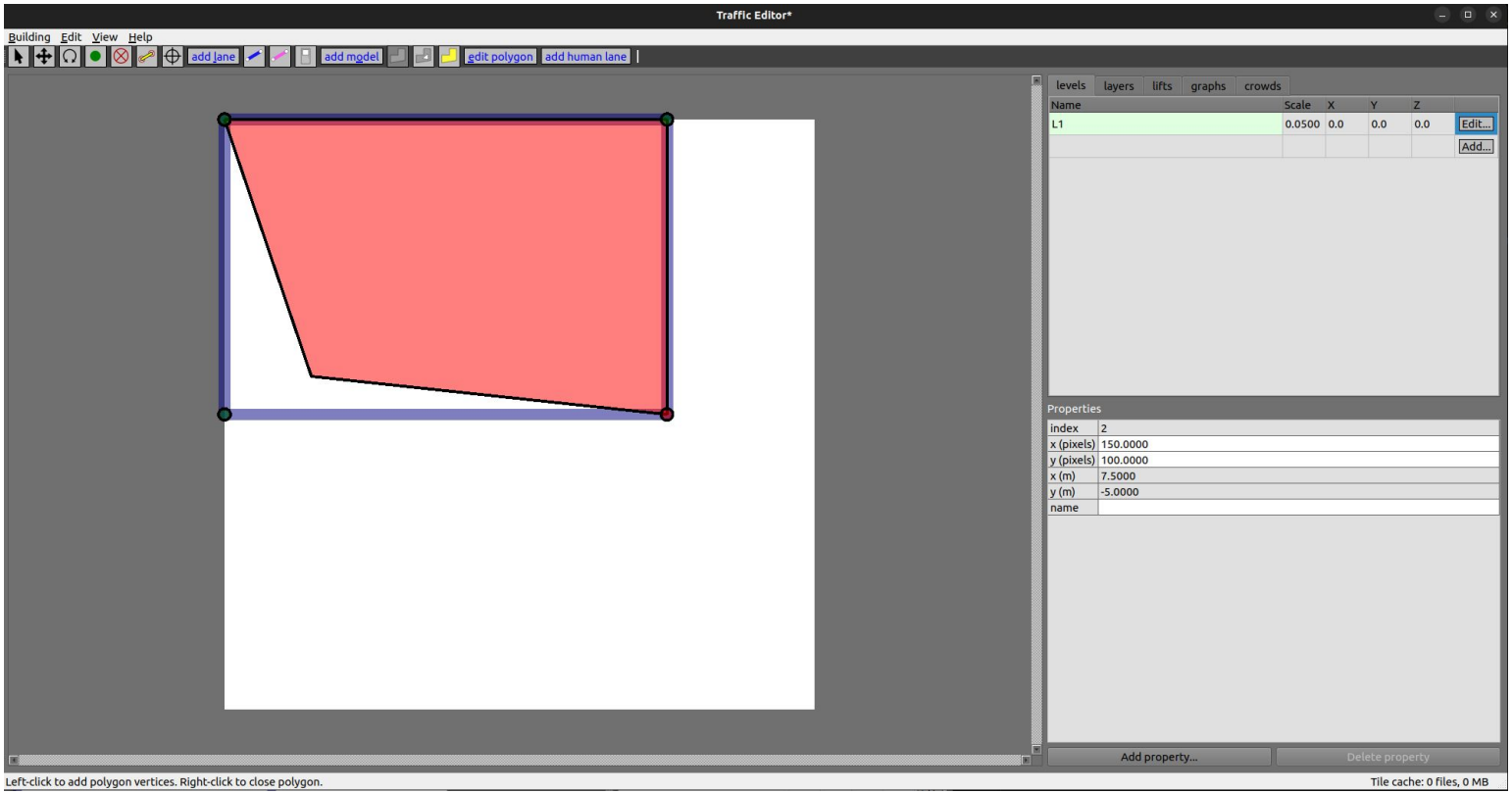
Add walls


Traffic Editor*
Building Edit View Help
add lane
add model
edit polygon
add human lane
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0500
0.0
0.0
0.0
Edit...
Add...
Properties
Add property...
Delete property
Click an item to select it.
Tile cache: 0 files, 0 MB

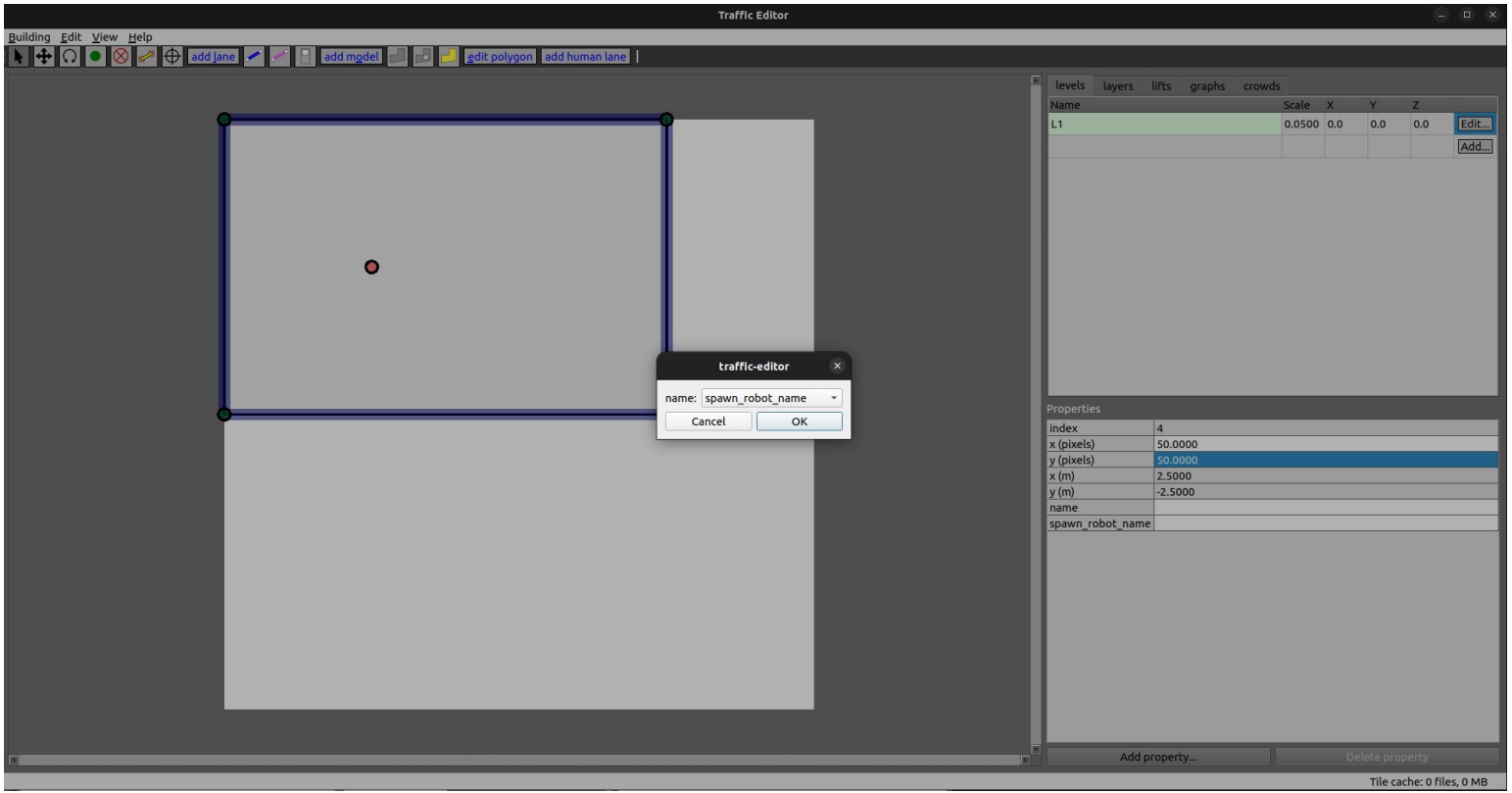
Align walls


Traffic Editor*
Building Edit View Help
add lane
add model
edit polygon
add human lane
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0500
0.0
0.0
0.0
Edit...
Add...
Properties
index
2
x (pixels)
150.0000
y (pixels)
100.0000
x (m)
7.5000
y (m)
-5.0000
name
Add property...
Delete property
Tile cache: 0 files, 0 MB

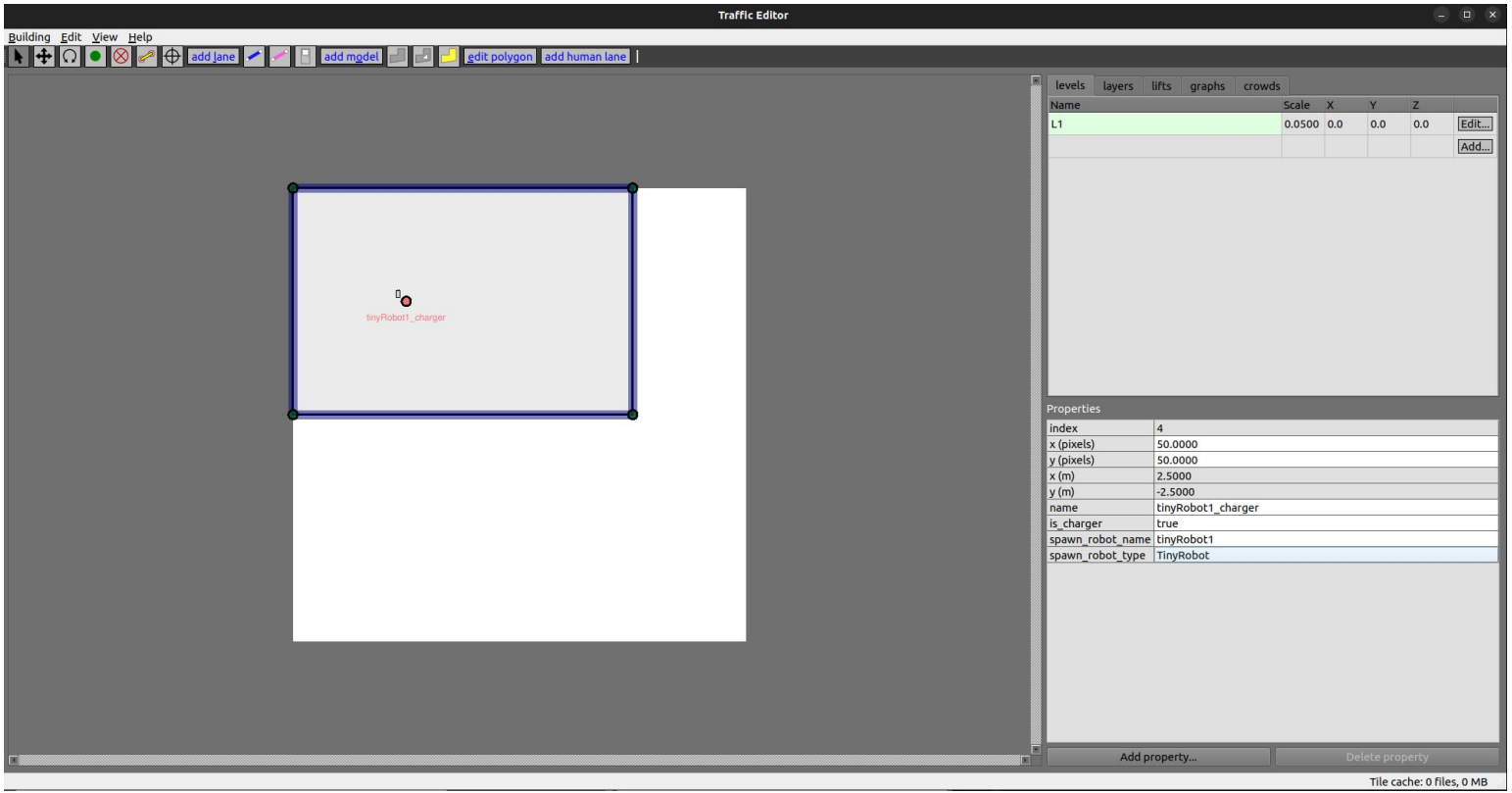
Add a floor


Traffic Editor*
Building Edit View Help
add lane
add model
edit polygon
add human lane
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0500
0.0
0.0
0.0
Edit...
Add...
Properties
index
2
x (pixels)
150.0000
y (pixels)
100.0000
x (m)
7.5000
y (m)
-5.0000
name
Add property...
Delete property
Left-click to add polygon vertices. Right-click to close polygon.
Tile cache: 0 files, 0 MB

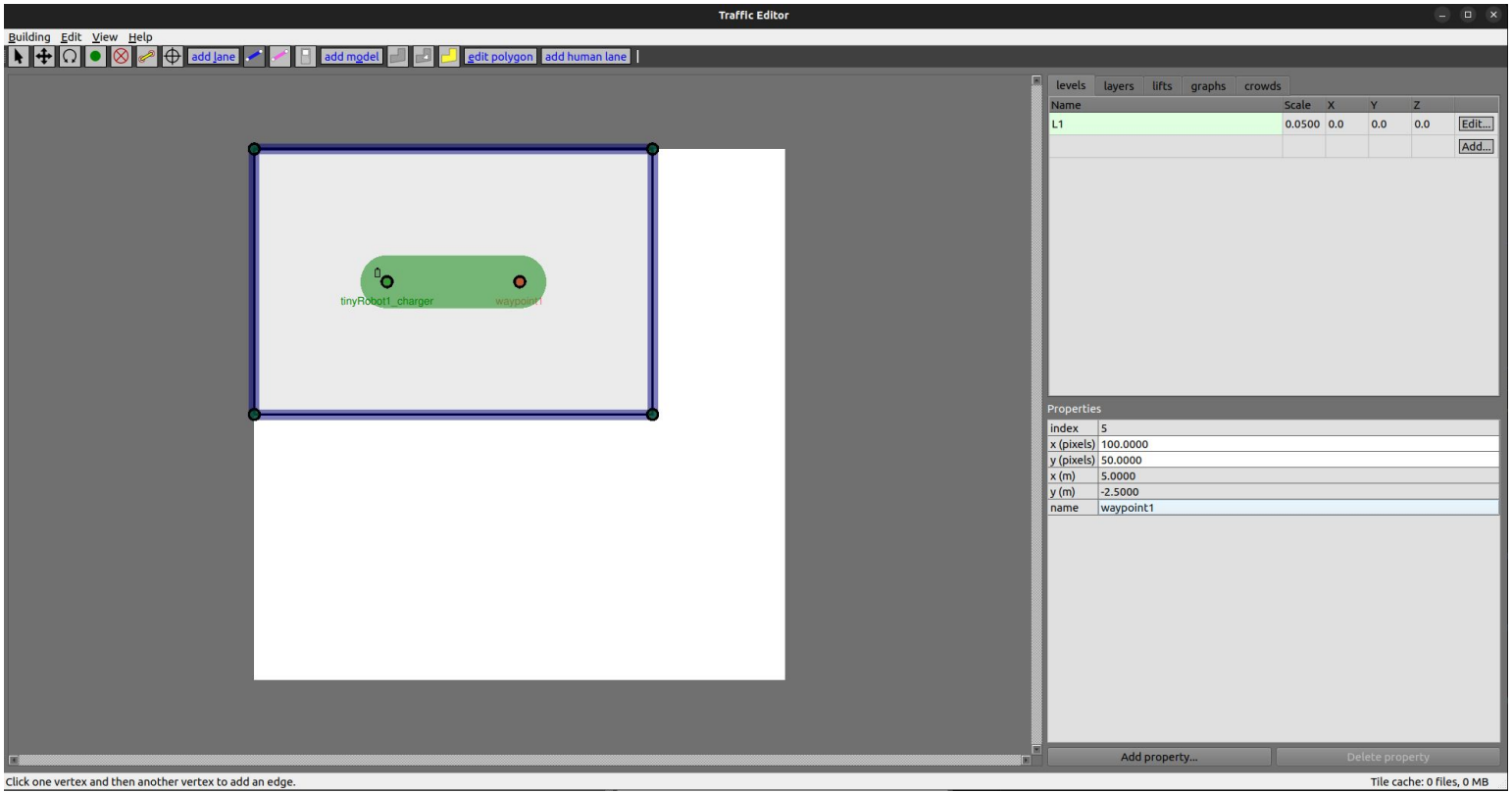
Spawn a robot



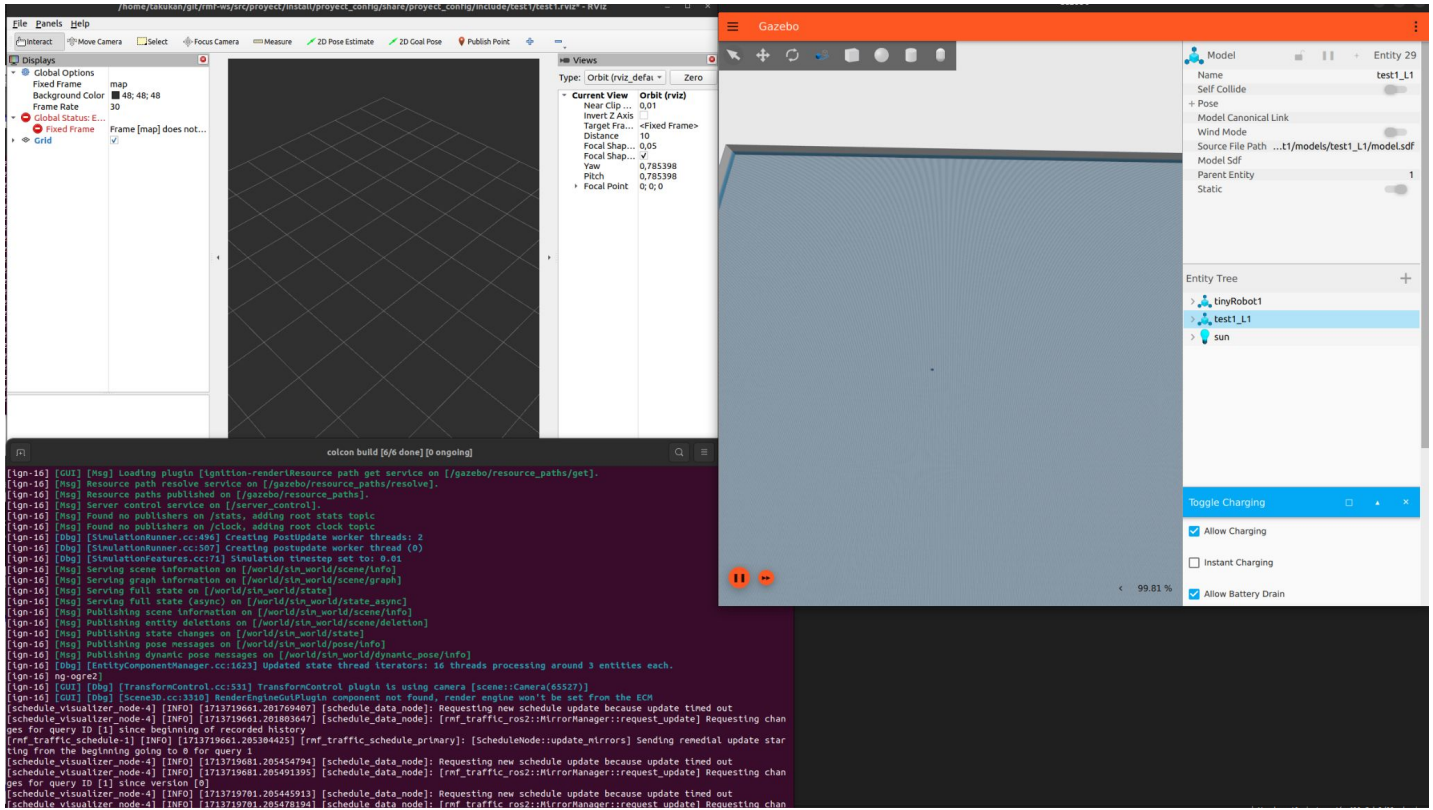
Robot configuration



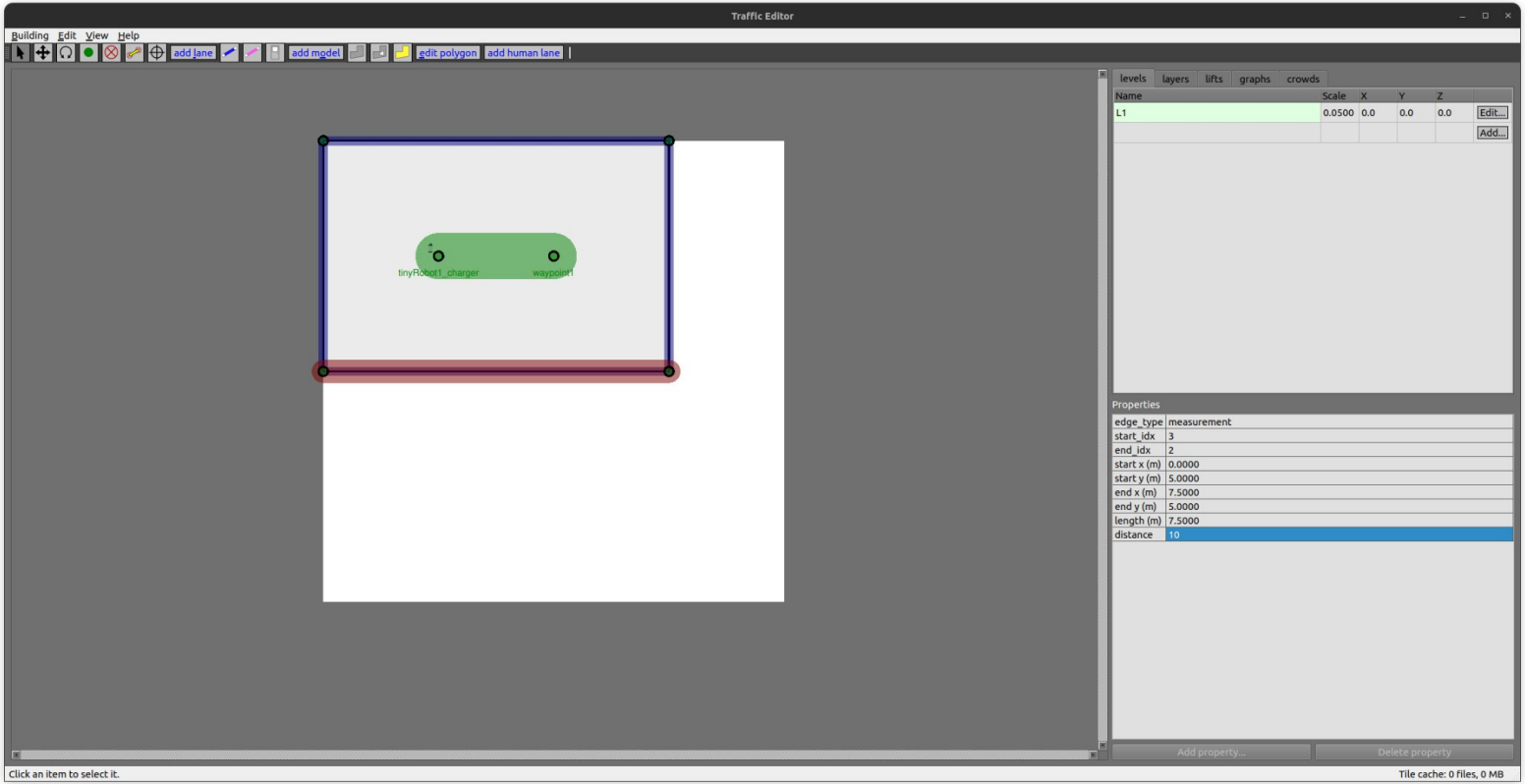
Add a lane


Traffic Editor
Building Edit View Help
add lane
add model
edit polygon
add human lane
tinyRobot1_charger
waypoint1
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0500
0.0
0.0
0.0
Edit...
Add...
Properties
index 5
x (pixels) 100.0000
y (pixels) 50.0000
x (m) 5.0000
y (m) -2.5000
name waypoint1
Add property...
Delete property
Click one vertex and then another vertex to add an edge.
Tile cache: 0 files, 0 MB

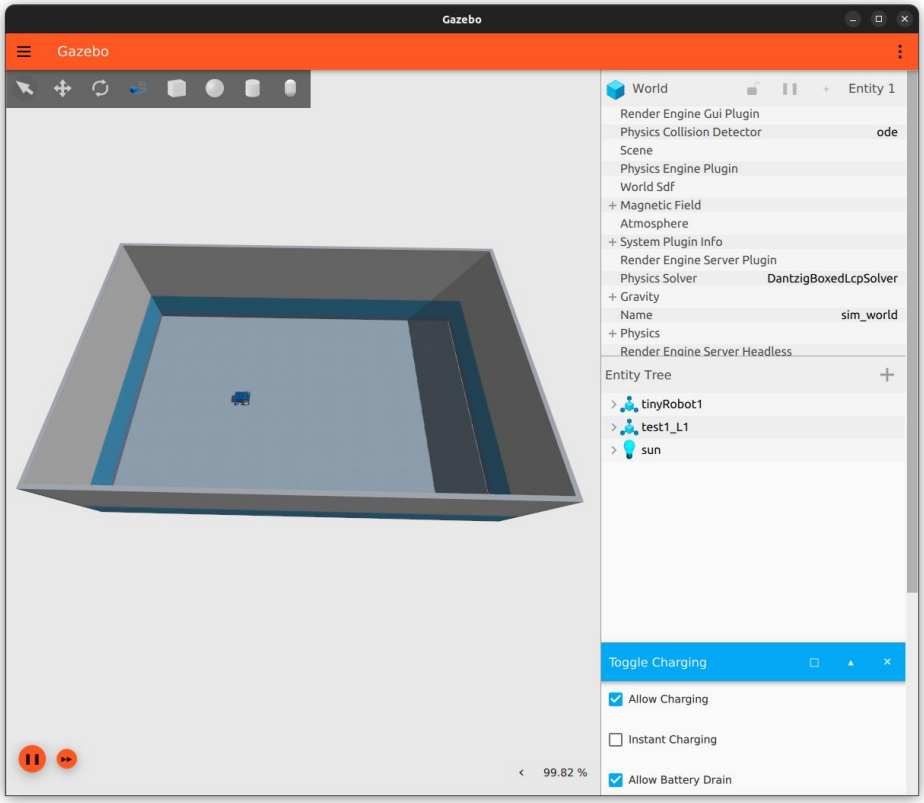
First launch



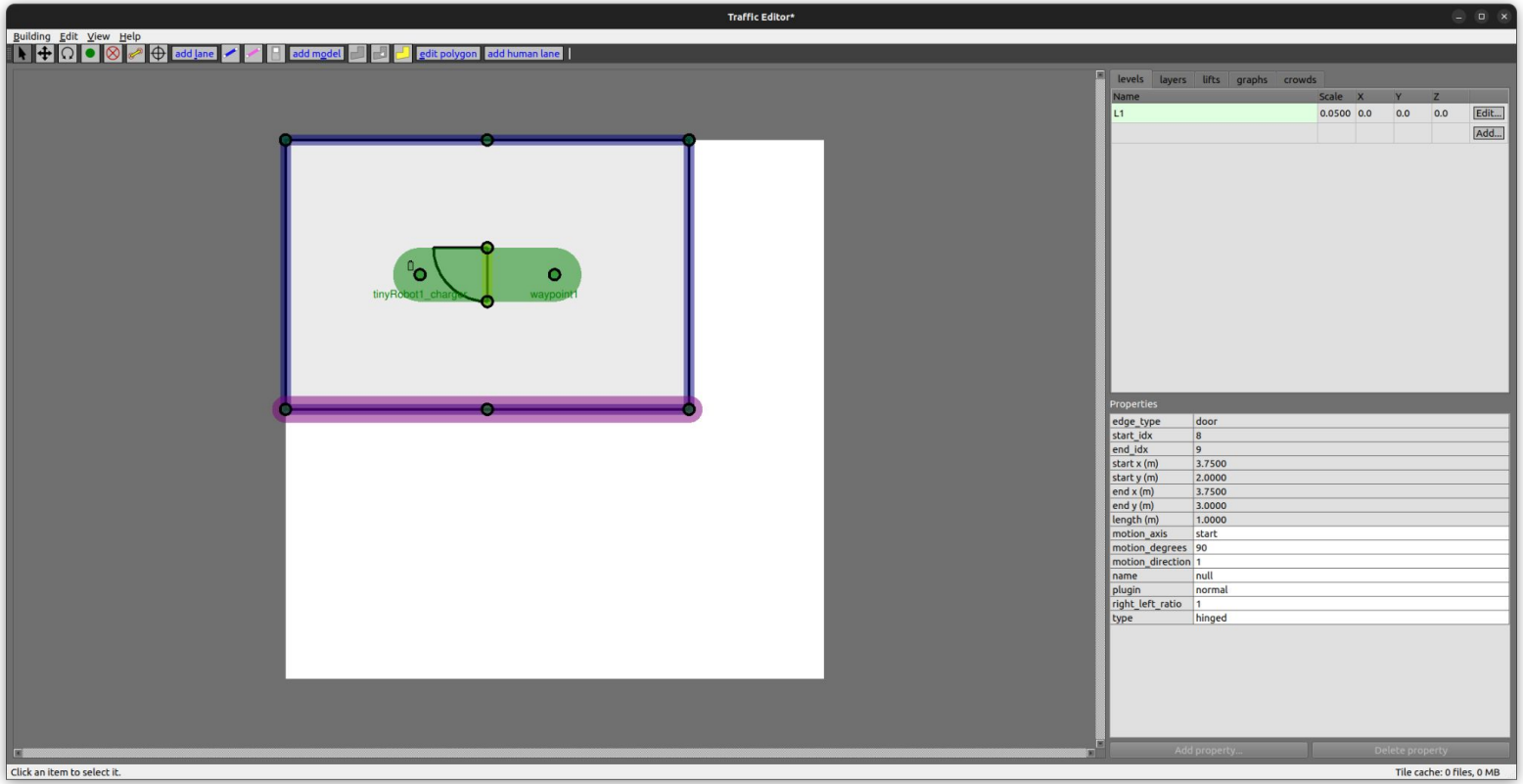
Set measurements



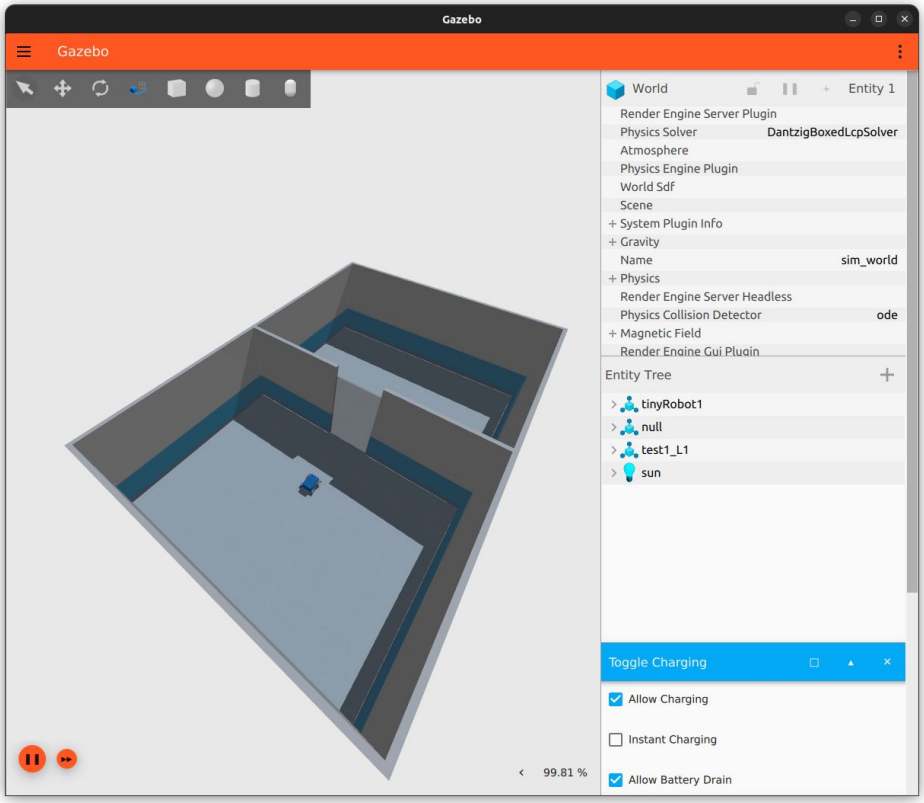
Measurement simulation



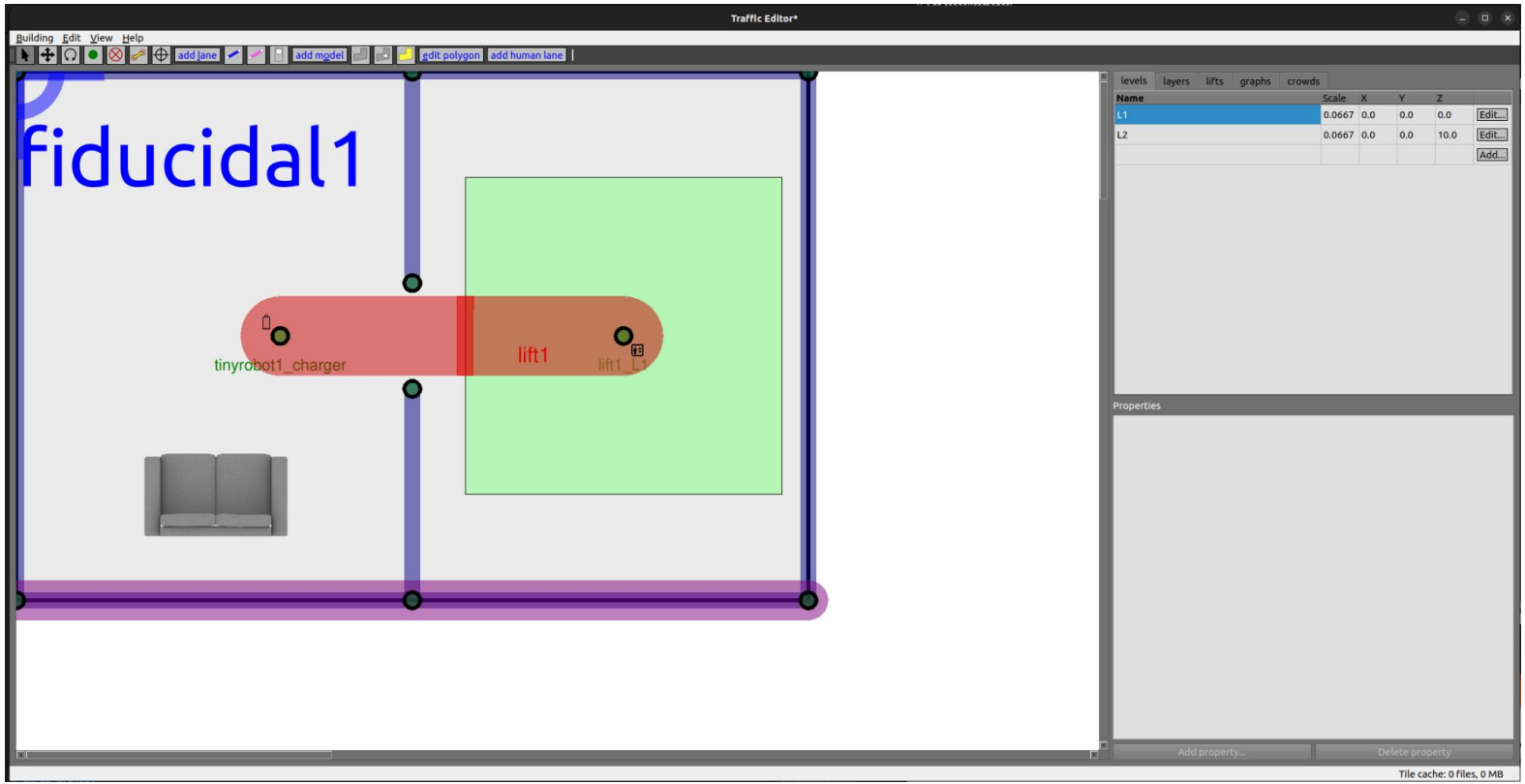
Add a door


Traffic Editor*
Building Edit View Help
add lane
add model
edit polygon
add human lane
tinyRobot1_charger
waypoint1
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0500
0.0
0.0
0.0
Edit...
Add...
Properties
edge_type door
start_idx 8
end_idx 9
start x (m) 3.7500
start y (m) 2.0000
end x (m) 3.7500
end y (m) 3.0000
length (m) 1.0000
motion_axis start
motion_degrees 90
motion_direction 1
name null
plugin normal
right_left_ratio 1
type hinged
Add property...
Delete property
Click an item to select it.
Tile cache: 0 files, 0 MB

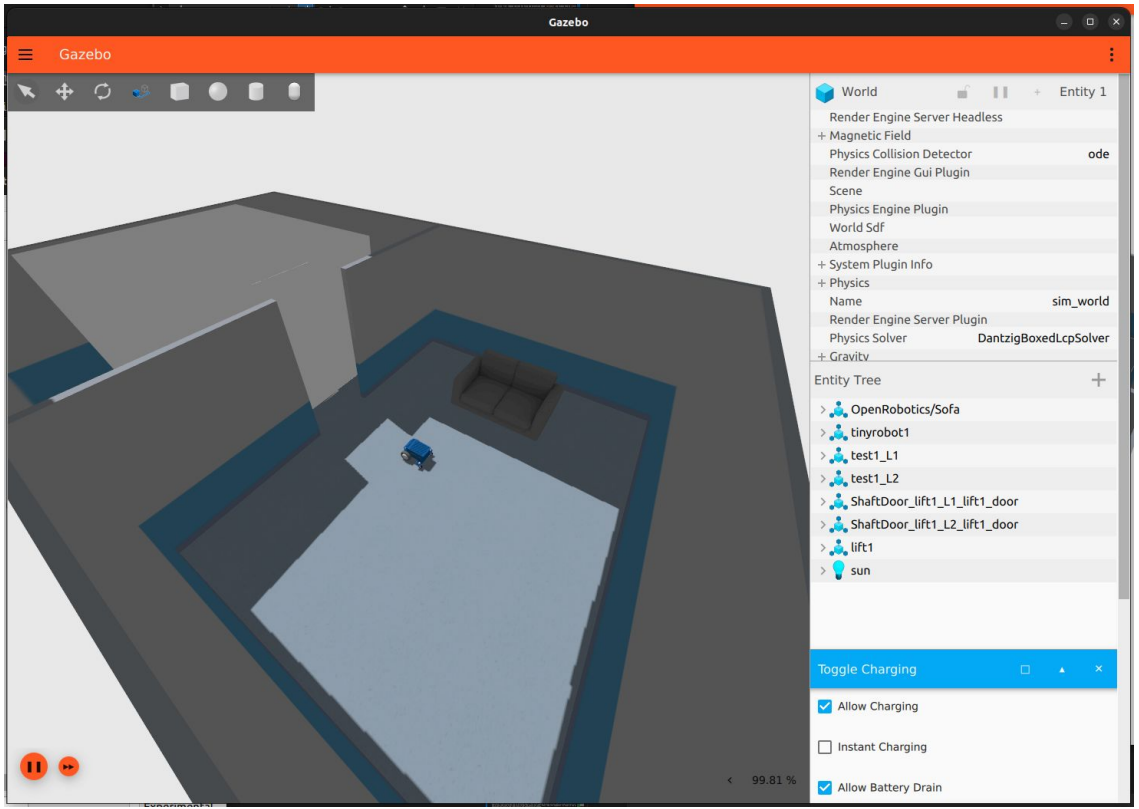
Door simulation



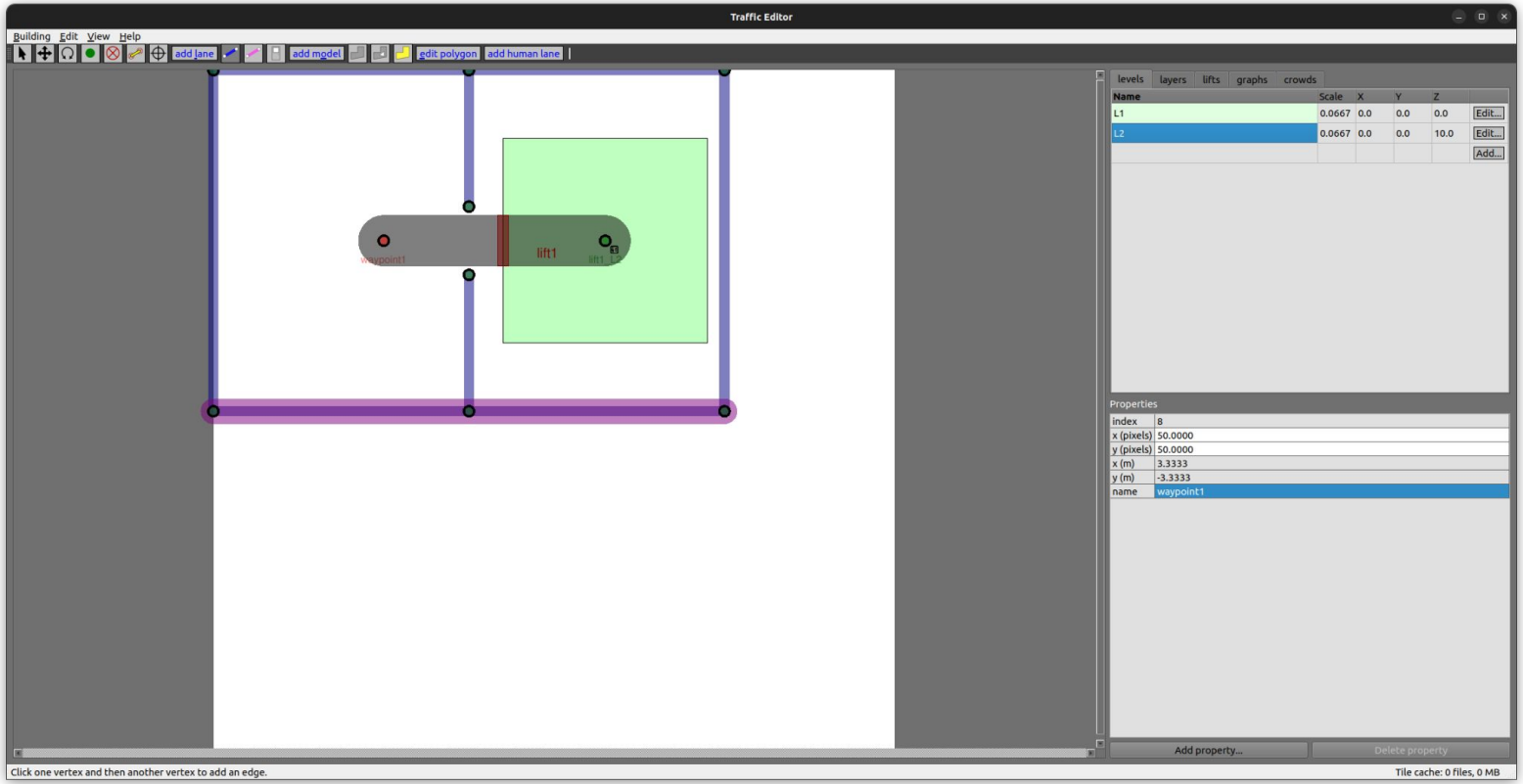
Add models


Traffic Editor*
Building Edit View Help
add lane
add model
edit polygon
add human lane
fiducidal1
tinyrobot1_charger
lift1
lift1_L1
levels
layers
lifts
graphs
crowds
Name
Scale
X
Y
Z
L1
0.0667
0.0
0.0
0.0
Edit...
L2
0.0667
0.0
0.0
10.0
Edit...
Add...
Properties
Add property...
Delete property
Tile cache: 0 files, 0 MB

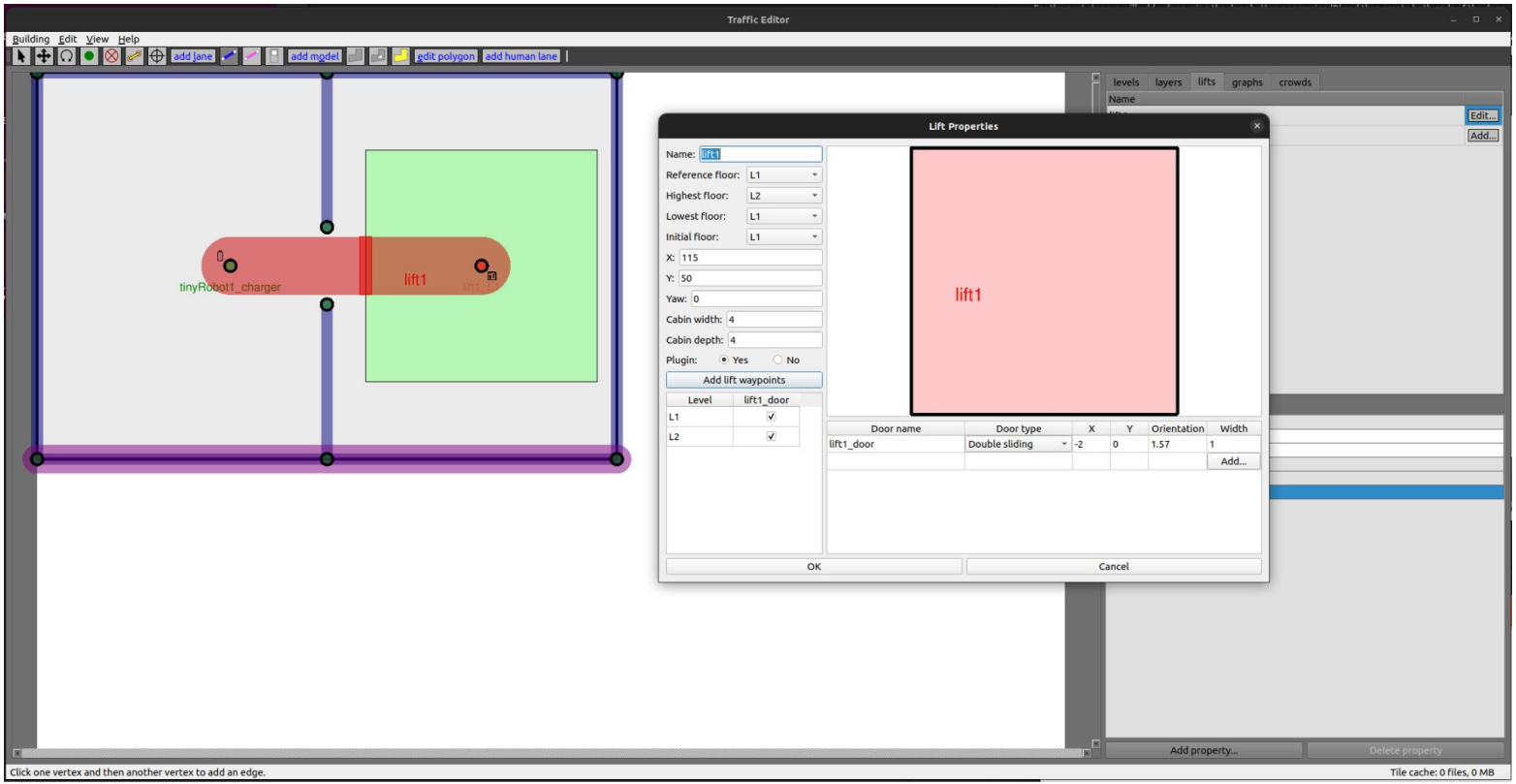
Simulation with models



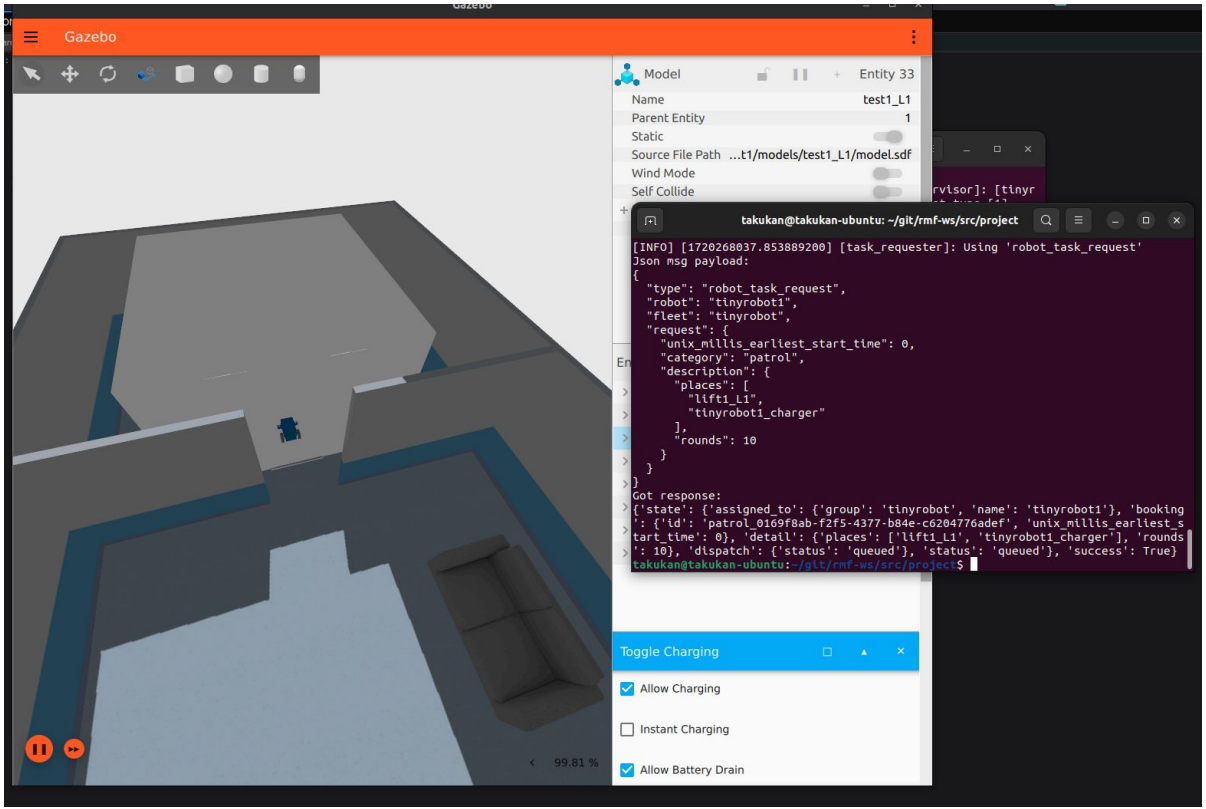
Add a second floor



Create a lift



Executing a task



Launch ICC_Kyoto world

```
rocker --nvidia --x11 \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  --network host --user \
  --volume `pwd`/rmf_ws:/home/usuario/rmf_ws --  \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest  \
    bash
```

Launch ICC_Kyoto world

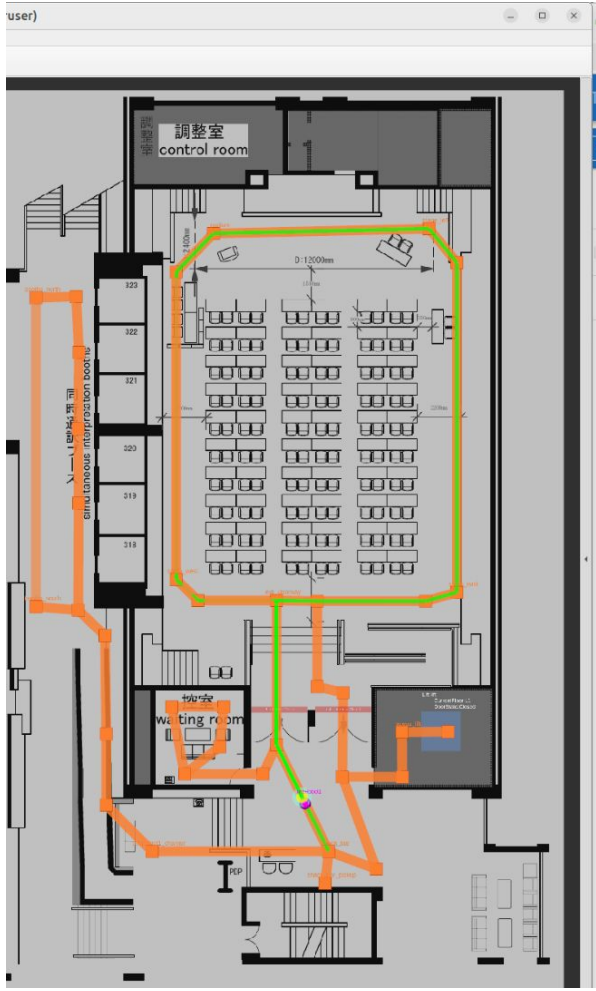
```
sudo cp -R /root/.gazebo .

cd rmf_ws/

source install/setup.bash

ros2 launch project_simulation \
  icc_kyoto.launch.xml \
  server_uri:="ws://localhost:8000/_internal"
```

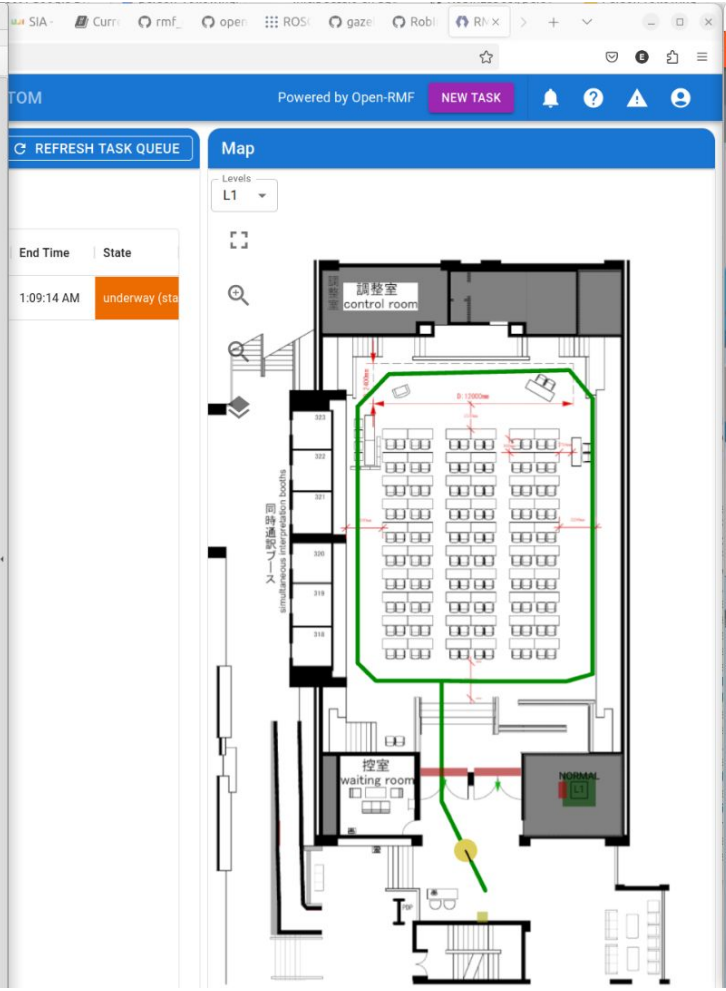
Terminal 2 - API Server

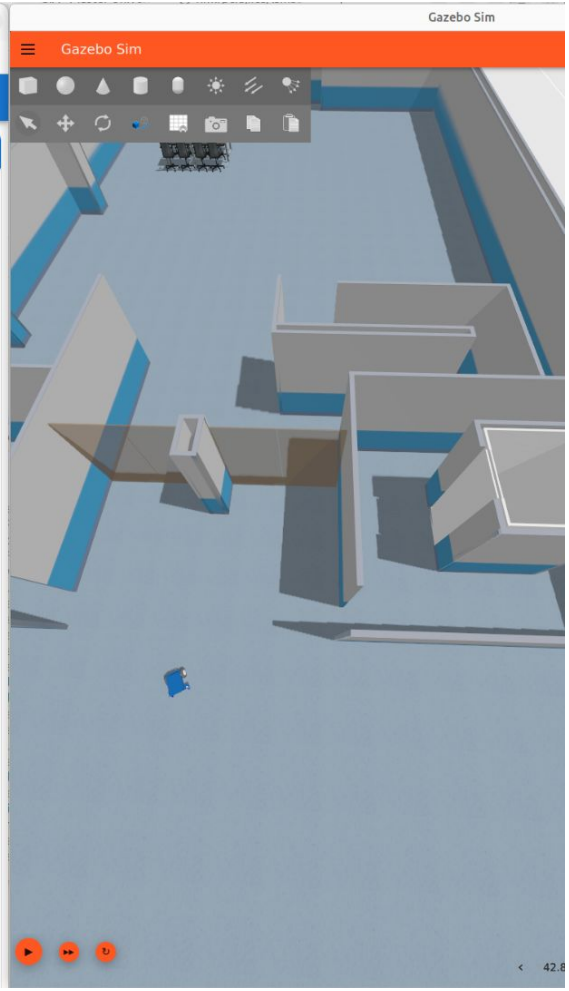
```
docker run --network host -it \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  -e RMW_IMPLEMENTATION=rmw_cyclonedds_cpp \
   ghcr.io/open-rmf/rmf-web/api-server:jazzy-nightly
```

Terminal 3 - dashboard

```
docker run --network host -it \
  -e RMF_SERVER_URL=http://localhost:8000 \
  -e TRAJECTORY_SERVER_URL=ws://localhost:8006 \
   ghcr.io/open-rmf/rmf-web/dashboard:jazzy-nightly
```


調整室
control room
simultaneous interpretation booths
waiting room


TOM
Powered by Open-RMF
NEW TASK
REFRESH TASK QUEUE
Map
Levels
L1
End Time
State
1:09:14 AM
underway (sta
調整室
control room
simultaneous interpretation booths
控室
waiting room
NORMAL


Gazebo Sim
Gazebo Sim

Exercise in GitHub repository

IMPORTANT: include a README in your repository with all the necessary commands for launching the simulations of the simple world and the ICC Kyoto world, and add some screen captures with examples of tasks.

Use the Open-RMF demos README as example.

References

Romero Mollá, J. (2024). Practical guide to implement OpenRMF in multi-robot environments. Treball Final de Grau en Enginyeria Informàtica. Escola Politècnica Superior de la Universitat d'Alacant.

RUA: Practical guide to implement OpenRMF in multi-robot environments